Problem Statement

- **Find all eigenvalues and eigenfunctions of the SLP:**

$$x^2y'' + 3xy' + \lambda y = 0, \qquad y(1) = 0,\ y(2) = 0.$$

- This is a **Cauchy–Euler equation**, so we try $y = x^m$.
- Substituting $y = x^m$ gives $y' = mx^{m-1}$, $y'' = m(m-1)x^{m-2}$, so

$$x^2\left[m(m-1)x^{m-2}\right] + 3x\left[mx^{m-1}\right] + \lambda x^m = 0.$$

- Dividing by x^m gives the **auxiliary (indicial) equation**

$$m(m-1) + 3m + \lambda = 0 \implies m^2 + 2m + \lambda = 0.$$

- Solving for m:

$$m = -1 \pm \sqrt{1-\lambda}.$$

- The nature of the two roots — real distinct, real repeated, or complex — depends on the sign of $1 - \lambda$, giving three cases.



Case I: $1 - \lambda > 0$ (i.e. $\lambda < 1$)

- The roots $m = -1 \pm \sqrt{1-\lambda}$ are real and distinct, so
$$y(x) = c_1 x^{-1+\sqrt{1-\lambda}} + c_2 x^{-1-\sqrt{1-\lambda}}.$$
- Apply $y(1) = 0$:
$$c_1 + c_2 = 0 \implies c_2 = -c_1.$$
- Hence
$$y(x) = c_1 \left(x^{-1+\sqrt{1-\lambda}} - x^{-1-\sqrt{1-\lambda}} \right).$$
- Apply $y(2) = 0$:
$$c_1 \left(2^{-1+\sqrt{1-\lambda}} - 2^{-1-\sqrt{1-\lambda}} \right) = 0.$$
- Since $\lambda < 1$ gives $\sqrt{1-\lambda} > 0$, the two powers of 2 are distinct, so the bracket is nonzero. Thus $c_1 = 0$, forcing $c_2 = 0$.
- **Conclusion:** $y(x) \equiv 0$; there is **no eigenvalue** with



Case II: $1 - \lambda = 0$ (i.e. $\lambda = 1$)

- Here $m = -1$ is a **repeated root**, so the general solution is
$$y(x) = c_1 x^{-1} + c_2 x^{-1} \ln x.$$
- Apply $y(1) = 0$: since $\ln 1 = 0$,
$$c_1 = 0.$$
- So $y(x) = c_2 x^{-1} \ln x$.
- Apply $y(2) = 0$:
$$\tfrac{1}{2} c_2 \ln 2 = 0 \implies c_2 = 0 \quad (\text{since } \ln 2 \neq 0).$$
- **Conclusion:** $y(x) \equiv 0$; there is **no eigenvalue** at $\lambda = 1$.



Case III: $1 - \lambda < 0$ (i.e. $\lambda > 1$) — Setting Up

- Now the roots are complex: $m = -1 \pm i\sqrt{\lambda - 1}$.
- The corresponding general (real) solution is

$$y(x) = x^{-1}\left[c_1 \cos\left(\sqrt{\lambda - 1}\,\ln x\right) + c_2 \sin\left(\sqrt{\lambda - 1}\,\ln x\right)\right].$$

- Apply $y(1) = 0$: since $\ln 1 = 0$, $\cos 0 = 1$, $\sin 0 = 0$,

$$c_1 = 0.$$

- Hence

$$y(x) = c_2\, x^{-1} \sin\left(\sqrt{\lambda - 1}\,\ln x\right).$$

- It remains to apply the second boundary condition $y(2) = 0$.



Case III — Eigenvalues and Eigenfunctions

- Apply $y(2) = 0$:

$$\tfrac{1}{2} c_2 \sin\left(\sqrt{\lambda - 1}\,\ln 2\right) = 0.$$

- For a **nontrivial** solution we need $c_2 \neq 0$, so

$$\sin\left(\sqrt{\lambda - 1}\,\ln 2\right) = 0 \implies \sqrt{\lambda - 1}\,\ln 2 = n\pi, \qquad n = 1, 2, 3, \ldots$$

- Solving for λ:

$$\sqrt{\lambda - 1} = \frac{n\pi}{\ln 2} \implies \boxed{\lambda_n = 1 + \frac{n^2\pi^2}{(\ln 2)^2}}, \qquad n = 1, 2, 3, \ldots$$

- The corresponding eigenfunctions are

$$\boxed{y_n(x) = x^{-1} \sin\left(\frac{n\pi \ln x}{\ln 2}\right)}, \qquad n = 1, 2, 3, \ldots$$

- So the Cauchy–Euler SLP has eigenvalues only for $\lambda > 1$, given by $\lambda_n = 1 + n^2\pi^2/(\ln 2)^2$.



Problem Statement

- **Find all eigenvalues and eigenfunctions of the SLP:**

$$\left[(2x+1)^{-1}y'\right]'+\lambda(2x+1)y=0, \qquad y(0)=0,\ y(\pi)=0.$$

- Writing $y' = dy/dx$ in terms of a new variable t, the equation becomes

$$\frac{d}{dx}\left[(2x+1)^{-1}\,\frac{dy/dt}{dx/dt}\right]+\lambda(2x+1)y=0.$$

- **Strategy:** choose the parametrization $t = t(x)$ so that the coefficient $(2x+1)^{-1}$ is absorbed, reducing the equation to one with constant coefficients.
- Put

$$\frac{dx}{dt}=(2x+1)^{-1},$$

which is well defined since $2x+1 \neq 0$ on $[0,\pi]$.



Finding the New Variable t

- From $\dfrac{dx}{dt} = (2x+1)^{-1}$ we get

$$(2x+1)\,dx = dt.$$

- Integrating, $t = x^2 + x$ (taking $t = 0$ at $x = 0$).
- With this parametrization, $\frac{dy}{dt}/\frac{dx}{dt} = (2x+1)\frac{dy}{dt}$ and the equation reduces to

$$\frac{d}{dx}\left[\frac{dy}{dt}\right] + \lambda(2x+1)y = 0.$$

- Since $\dfrac{dt}{dx} = 2x+1$, the chain rule gives

$$\frac{d}{dx}\left[\frac{dy}{dt}\right] = \frac{d^2y}{dt^2}\cdot\frac{dt}{dx} = (2x+1)\frac{d^2y}{dt^2}.$$

- Substituting and dividing throughout by $(2x+1) \neq 0$:

$$\boxed{\frac{d^2y}{dt^2} + \lambda y = 0}.$$



Transformed Boundary Value Problem

- Under $t = x^2 + x$:
  - at $x = 0$: $t = 0$;
  - at $x = \pi$: $t = \pi^2 + \pi = \pi(\pi + 1)$.
- So the problem becomes the **constant-coefficient** SLP

$$\frac{d^2y}{dt^2} + \lambda y = 0, \qquad y(0) = 0,\ y\left(\pi^2 + \pi\right) = 0.$$

- This is the classical eigenvalue problem for the operator $-d^2/dt^2$ on the interval $[0, \pi^2 + \pi]$ with Dirichlet conditions.
- As usual, we examine the three cases $\lambda = 0$, $\lambda < 0$, $\lambda > 0$ in turn.



Case $\lambda = 0$ and Case $\lambda < 0$

- **Case $\lambda = 0$:** the equation is $y'' = 0$, so $y(t) = c_1 + c_2 t$.
  - $y(0) = 0 \Rightarrow c_1 = 0$.
  - $y(\pi^2 + \pi) = 0 \Rightarrow c_2(\pi^2 + \pi) = 0 \Rightarrow c_2 = 0$.

  Hence $y \equiv 0$: **no eigenvalue** at $\lambda = 0$.
- **Case $\lambda < 0$:** write $\lambda = -p^2$, $p > 0$. Then $y'' - p^2 y = 0$, so

$$y(t) = c_1 \cosh(pt) + c_2 \sinh(pt).$$

  - $y(0) = 0 \Rightarrow c_1 = 0$.
  - $y(\pi^2 + \pi) = 0 \Rightarrow c_2 \sinh\left(p(\pi^2 + \pi)\right) = 0$. Since sinh vanishes only at 0 and $p(\pi^2 + \pi) \neq 0$, we get $c_2 = 0$.

  Hence $y \equiv 0$: **no eigenvalue** for $\lambda < 0$.



Case $\lambda > 0$ — Eigenvalues and Eigenfunctions

- For $\lambda > 0$, the general solution is
$$y(t) = c_1 \cos\sqrt{\lambda}\, t + c_2 \sin\sqrt{\lambda}\, t.$$
- Apply $y(0) = 0$: $c_1 = 0$, so $y(t) = c_2 \sin\sqrt{\lambda}\, t$.
- Apply $y(\pi^2 + \pi) = 0$, i.e. $y\big(\pi(\pi+1)\big) = 0$. For a nontrivial solution,
$$\sin\big(\sqrt{\lambda}\,\pi(\pi+1)\big) = 0 \implies \sqrt{\lambda}\,\pi(\pi+1) = n\pi, \qquad n = 1, 2, 3, .$$
- The factor of π cancels:
$$\sqrt{\lambda}\,(\pi+1) = n \implies \boxed{\lambda_n = \frac{n^2}{(\pi+1)^2}}, \qquad n = 1, 2, 3, \ldots$$
- Returning to the original variable $t = x^2 + x$, the eigenfunctions are
$$\boxed{y_n(x) = \sin\left(\frac{n(x^2+x)}{\pi+1}\right)}, \qquad n = 1, 2, 3, \ldots$$



Orthogonality Theorem — Statement

- **Theorem.** Let $y_m(x), y_n(x)$ be eigenfunctions of a S–L problem for distinct eigenvalues $\lambda_m \neq \lambda_n$.
- Then y_m and y_n are **orthogonal** with respect to the weight $w(x)$ on $[a, b]$:

$$\int_a^b w(x)\, y_m(x)\, y_n(x)\, dx = 0$$

- This holds for regular, singular, and periodic S–L problems, provided the boundary terms vanish.
- This is the direct generalization of orthogonality of $\sin(n\pi x/L)$ in classical Fourier series.
- It is the foundation for eigenfunction (generalized Fourier) expansions.
- Distinct choices of $w(x)$ give distinct families: Legendre polynomials ($w = 1$), Bessel functions ($w = x$), Hermite functions ($w = e^{-x^2}$), etc.



Orthogonality Theorem — Proof (I)

- y_m, y_n satisfy $(py_m')' + (q + \lambda_m w)y_m = 0$ and $(py_n')' + (q + \lambda_n w)y_n = 0$.
- Multiply the first by y_n, the second by y_m, and subtract:

$$y_n(py_m')' - y_m(py_n')' + (\lambda_m - \lambda_n) w\, y_m y_n = 0$$

- Note that $y_n(py_m')' - y_m(py_n')' = \dfrac{d}{dx}\left[p(y_m' y_n - y_n' y_m)\right]$ (Lagrange's identity).
- Substituting and integrating over $[a, b]$:

$$\left[p(y_m' y_n - y_n' y_m)\right]_a^b + (\lambda_m - \lambda_n) \int_a^b w\, y_m y_n\, dx = 0$$



Orthogonality Theorem — Proof (II)

- The boundary conditions (separated, periodic, or singular-with-finiteness) force the boundary term to vanish:
$$\left[p(y_m' y_n - y_n' y_m)\right]_a^b = 0$$
- For separated BCs: at $x = a$, $\alpha_1 y_m(a) + \alpha_2 y_m'(a) = 0$ and similarly for y_n force the Wronskian-type term to vanish (both satisfy the same linear relation).
- For singular endpoints with $p(a) = 0$ or $p(b) = 0$: the term vanishes automatically since p vanishes there.
- For periodic BCs: the term at a cancels the term at b since p, y, y' match at both ends.
- Hence $(\lambda_m - \lambda_n) \int_a^b w\, y_m y_n\, dx = 0$.
- Since $\lambda_m \neq \lambda_n$, we conclude $\int_a^b w\, y_m y_n\, dx = 0$. ■



Weighted Inner Product

- Define the weighted inner product of two functions u, v on $[a, b]$:

$$\langle u, v \rangle_w = \int_a^b w(x)\, u(x)\, v(x)\, dx$$

- Orthogonality of eigenfunctions reads simply as $\langle y_m, y_n \rangle_w = 0$ for $m \neq n$.
- This inner product makes the eigenfunctions $\{y_n\}$ into an orthogonal system in the space $L^2_w[a, b]$.
- The weight $w(x)$ is dictated entirely by the coefficient of λ in the S–L equation.
- Different S–L equations give rise to different natural inner products for expansions.
- This is analogous to the dot product $\mathbf{u} \cdot \mathbf{v}$ for orthogonal eigenvectors of a symmetric matrix.



Norm of an Eigenfunction

- The (weighted) norm of an eigenfunction y_n is

$$\|y_n\|_w = \left(\int_a^b w(x) \, [y_n(x)]^2 \, dx \right)^{1/2}$$

- This quantity is always finite and strictly positive for a nontrivial eigenfunction.
- Norms are used to normalize eigenfunctions for use in expansions.
- Explicit norm formulas are known for the classical families: e.g. Legendre polynomials, Fourier–Bessel functions.
- The norm generally depends on n (or on the eigenvalue λ_n).
- Computing $\|y_n\|_w$ is often the most computational step of an eigenfunction-expansion problem.



Normalized Eigenfunctions

- Define $\phi_n(x) = \dfrac{y_n(x)}{\|y_n\|_w}$.
- Then $\{\phi_n\}$ is an **orthonormal** system: $\langle \phi_m, \phi_n \rangle_w = \delta_{mn}$ (Kronecker delta).
- Orthonormal eigenfunctions simplify the coefficient formula in eigenfunction expansions.
- This is the functional analogue of normalizing eigenvectors to unit length.
- Example: $\phi_n(x) = \sqrt{2/L}\,\sin(n\pi x/L)$ for the classical Dirichlet problem on $[0, L]$.
- Normalized eigenfunctions are used directly in Parseval-type identities.



Eigenfunction Expansion Theorem

- **Theorem.** Let $\{y_n\}$ be the eigenfunctions of a regular S–L problem on $[a, b]$.
- Any piecewise smooth function $f(x)$ on $[a, b]$ can be expanded as a **generalized Fourier series**:

$$f(x) \sim \sum_{n=1}^{\infty} c_n\, y_n(x)$$

- The coefficients are given by

$$c_n = \frac{\langle f, y_n \rangle_w}{\langle y_n, y_n \rangle_w} = \frac{\int_a^b w(x) f(x) y_n(x)\, dx}{\int_a^b w(x)[y_n(x)]^2\, dx}$$

- This generalizes ordinary Fourier sine/cosine series (which is the special case $y_n = \sin(n\pi x/L)$ or $\cos(n\pi x/L)$).
- Legendre series, Fourier–Bessel series, Hermite and Laguerre series are all special cases.



Orthogonality for Cauchy–Euler S–L Problems

- For $x^2y'' + xy' + \lambda y = 0$ on $[1, e]$, with boundary conditions $y(1) = 0,\ y(e) = 0$, the weight function is $w(x) = 1/x$.
- Orthogonality: for $m \neq n$,

$$\int_1^e \frac{1}{x} \sin(m\pi \ln x) \sin(n\pi \ln x)\, dx = 0$$

- Verify via substitution $t = \ln x$, $dt = dx/x$: the integral becomes $\int_0^1 \sin(m\pi t) \sin(n\pi t)\, dt = 0$ for $m \neq n$ — the classical trigonometric orthogonality.
- Norm: $\int_1^e \frac{1}{x} \sin^2(n\pi \ln x)\, dx = \int_0^1 \sin^2(n\pi t)\, dt = \frac{1}{2}$.
- This confirms the general orthogonality theorem in this specific case, with weight $w = 1/x$.
- Any function $f(x)$ on $[1, e]$ can be expanded as $f(x) \sim \sum_n c_n \sin(n\pi \ln x)$, $c_n = 2 \int_1^e \frac{f(x)}{x} \sin(n\pi \ln x)\, dx$.



Legendre's Equation as an S–L Problem

- Legendre's equation: $(1-x^2)y'' - 2xy' + n(n+1)y = 0$, $-1 \le x \le 1$.
- Already in self-adjoint form: $\left[(1-x^2)y'\right]' + n(n+1)y = 0$.
- Identify $p(x) = 1-x^2,\ q(x) = 0,\ w(x) = 1,\ \lambda = n(n+1)$.
- Singular at $x = \pm 1$, since $p(\pm 1) = 0$: a **singular** S–L problem.
- Boundedness of y at $x = \pm 1$ (rather than a boundary condition) selects the eigenvalues $\lambda_n = n(n+1)$, $n = 0, 1, 2, \ldots$
- Eigenfunctions are the **Legendre polynomials** $P_n(x)$.



Problem

- Find the eigenvalues and eigenfunctions of the S–L problem

$$y'' + \lambda y = 0, \qquad y(0) = 0,\ y(\pi) = 0$$

- Verify directly (by integration) that the eigenfunctions are orthogonal on $[0, \pi]$.
- By Case A with $L = \pi$: eigenvalues $\lambda_n = n^2$, $n = 1, 2, 3, \ldots$
- Eigenfunctions: $y_n(x) = \sin(nx)$.
- Orthogonality check: $\displaystyle\int_0^\pi \sin(mx)\sin(nx)\,dx =$ $\frac{1}{2}\displaystyle\int_0^\pi \big[\cos((m-n)x) - \cos((m+n)x)\big]\,dx = 0$ for $m \neq n$ integers.
- Norm: $\displaystyle\int_0^\pi \sin^2(nx)\,dx = \pi/2.$



Problem

- Find the eigenvalues and eigenfunctions of

$$y'' + \lambda y = 0, \qquad y'(0) = 0,\ y'(\pi) = 0$$

- State the corresponding weight function and write the orthogonality relation explicitly.
- By Case B with $L = \pi$: eigenvalues $\lambda_n = n^2$, $n = 0, 1, 2, \ldots$
- Eigenfunctions: $y_n(x) = \cos(nx)$.
- Weight function $w(x) = 1$; orthogonality:
$$\int_0^{\pi} \cos(mx)\cos(nx)\,dx = 0,\ m \neq n.$$
- Norms: $\|y_0\|^2 = \pi$; $\|y_n\|^2 = \pi/2$ for $n \geq 1$.



Problem

- Find the eigenvalues and eigenfunctions of

$$y'' + \lambda y = 0, \qquad y(0) = 0,\ y'(1) = 0$$

- Sketch the first three eigenfunctions and count their interior zeros.
- By Case C with $L = 1$: $\lambda_n = \left(\dfrac{(2n-1)\pi}{2}\right)^2$, $n = 1, 2, 3, \ldots$
- Eigenfunctions: $y_n(x) = \sin\left(\dfrac{(2n-1)\pi x}{2}\right)$.
- $y_1(x) = \sin(\pi x/2)$ has 0 interior zeros on $(0, 1)$; $y_2(x) = \sin(3\pi x/2)$ has 1 interior zero; y_3 has 2.



Problem

- (Cauchy–Euler variety) Solve the S–L problem

$$x^2y'' + xy' + \lambda y = 0, \qquad y(1) = 0,\ y(e) = 0$$

- Find all eigenvalues and eigenfunctions, and write the orthogonality relation with the correct weight function.
- Here eigenvalues $\lambda_n = (n\pi)^2$, eigenfunctions $y_n(x) = \sin(n\pi \ln x)$, $n = 1, 2, 3, \ldots$ (HOMEWORK).
- Weight function $w(x) = 1/x$ (from the self-adjoint form $(xy')' + \lambda y/x = 0$).
- Orthogonality: $\displaystyle\int_1^e \frac{1}{x} \sin(m\pi \ln x) \sin(n\pi \ln x)\, dx = 0$, $m \neq n$.
- Verified via $t = \ln x$, reducing to classical trigonometric orthogonality on $[0, 1]$.



Problem

- Let y_m, y_n be eigenfunctions of a regular S–L problem $(py')' + (q + \lambda w)y = 0$ on $[a, b]$ with separated boundary conditions, for distinct eigenvalues $\lambda_m \neq \lambda_n$.
- Prove directly, from Lagrange's identity, that
$$\int_a^b w\, y_m y_n\, dx = 0.$$
- Multiply the equation for y_m by y_n and the equation for y_n by y_m; subtract, using Lagrange's identity.
- Integrate from a to b to get
$$\left[p(y_m' y_n - y_n' y_m)\right]_a^b + (\lambda_m - \lambda_n) \int_a^b w\, y_m y_n\, dx = 0$$
- The separated boundary conditions make the bracketed boundary term vanish at both $x = a$ and $x = b$.
- Since $\lambda_m \neq \lambda_n$, dividing through gives $\int_a^b w\, y_m y_n\, dx = 0.$



Problem

- Reduce the general equation

$$y'' + \frac{2}{x}y' + \lambda y = 0, \qquad x > 0$$

- to self-adjoint Sturm–Liouville form, by finding the appropriate integrating factor $p(x)$ and identifying $p(x), q(x), w(x)$.
- Here $a_0 = 1,\ a_1 = 2/x,\ a_2 = 0,\ a_3 = 1.$
- Integrating factor: $p(x) = \exp\left(\int \frac{2}{x}\,dx\right) = e^{2\ln x} = x^2.$
- Multiply the equation by x^2: $x^2y'' + 2xy' + \lambda x^2 y = 0.$
- This is $(x^2y')' + \lambda x^2 y = 0$: so $p(x) = x^2,\ q(x) = 0,\ w(x) = x^2.$
- (This is the radial part of the 3-D Laplacian, related to spherical Bessel functions.)



Problem

- Reduce Chebyshev's equation

$$(1-x^2)y'' - xy' + n^2 y = 0, \qquad -1 < x < 1$$

- to self-adjoint form, and identify the weight function $w(x)$ used in the orthogonality relation for Chebyshev polynomials $T_n(x)$.
- Divide by $(1-x^2)$: $y'' - \dfrac{x}{1-x^2}\,y' + \dfrac{n^2}{1-x^2}y = 0.$
- Integrating factor:

$$p(x) = \exp\left(-\int \frac{x}{1-x^2}dx\right) = (1-x^2)^{1/2}.$$

- Multiplying: $\sqrt{1-x^2}\,y'' - \dfrac{x}{\sqrt{1-x^2}}y' + \dfrac{n^2}{\sqrt{1-x^2}}y = 0,$ i.e.

$$\left[\sqrt{1-x^2}\,y'\right]' + \frac{n^2}{\sqrt{1-x^2}}\,y = 0$$



Problem

- Find all eigenvalues and eigenfunctions of the periodic S–L problem

$$y'' + \lambda y = 0, \qquad y(-\pi) = y(\pi),\ y'(-\pi) = y'(\pi)$$

  State which eigenvalues have multiplicity greater than one.
- By Case E with $L = \pi$: eigenvalues $\lambda_n = n^2$, $n = 0, 1, 2, \ldots$
- $n = 0$: eigenfunction $y_0 = 1$ (multiplicity 1).
- $n \geq 1$: eigenfunctions $\cos(nx)$ and $\sin(nx)$ (multiplicity 2 for each $\lambda_n = n^2$, $n \geq 1$).
- This is the classical full-range Fourier series eigenvalue problem.



Problem

- (Robin case) Find the equation satisfied by the eigenvalues of

$$y'' + \lambda y = 0, \qquad y(0) = 0, \quad y'(1) + 2y(1) = 0$$

  Is this equation solvable in closed form? Explain.
- As in Case F with $L = 1,\ h = 2$: with $\lambda = p^2$, $y = c_2 \sin(px)$ (from $y(0) = 0$).
- Condition: $y'(1) + 2y(1) = c_2\big[p + 2\big] = 0.$
- Transcendental equation: $= -p/2.$
- Not solvable in closed form; roots $p_1 < p_2 < \cdots$ are found graphically/numerically, then $\lambda_n = p_n^2$.



Problem

- Expand $f(x) = 1$ (constant function) on $[0, L]$ in the eigenfunctions of the Dirichlet problem

$$y'' + \lambda y = 0, \qquad y(0) = 0,\ y(L) = 0$$

. Find the generalized Fourier coefficients c_n explicitly.
- Eigenfunctions: $y_n(x) = \sin(n\pi x/L)$; weight $w = 1$; $\text{norm}^2 = L/2$.
- Coefficient formula: $c_n = \dfrac{2}{L}\displaystyle\int_0^L \sin\left(\frac{n\pi x}{L}\right) dx.$
- Evaluate:
$$\int_0^L \sin\left(\frac{n\pi x}{L}\right) dx = \frac{L}{n\pi}\left[1 - \cos(n\pi)\right] = \frac{L}{n\pi}\left[1 - (-1)^n\right].$$
- So $c_n = \dfrac{2}{n\pi}\left[1 - (-1)^n\right]$: $c_n = \dfrac{4}{n\pi}$ for n odd, $c_n = 0$ for n even.
- Hence $1 \sim \displaystyle\sum \frac{4}{n\pi} \sin\left(\frac{n\pi x}{L}\right)$ on $(0, L)$.

